Cubic & Cube Root Inverses — Practice

Name: ____________________ Period: ______

Part 1: Find the inverse algebraically. Use the 4-step method. Show your work.

1. $f(x) = x^3 + 7$

$f^{-1}(x) =$ ______________________

2. $f(x) = \sqrt[3]{x} - 3$

$f^{-1}(x) =$ ______________________

3. $f(x) = 2x^3$

$f^{-1}(x) =$ ______________________

4. $f(x) = \sqrt[3]{x+5}$

$f^{-1}(x) =$ ______________________

5. $f(x) = -x^3 + 4$

$f^{-1}(x) =$ ______________________

6. $f(x) = 3\sqrt[3]{x}$

$f^{-1}(x) =$ ______________________

7. $f(x) = \frac{1}{2}x^3 - 1$

$f^{-1}(x) =$ ______________________

8. $f(x) = -\sqrt[3]{x} + 2$

$f^{-1}(x) =$ ______________________

Part 2: Graph the function and its inverse. Complete the table, find the inverse, swap points, and graph both.

9. $f(x) = x^3 - 3$ *(Cubic shifted down 3)*

Find the inverse:

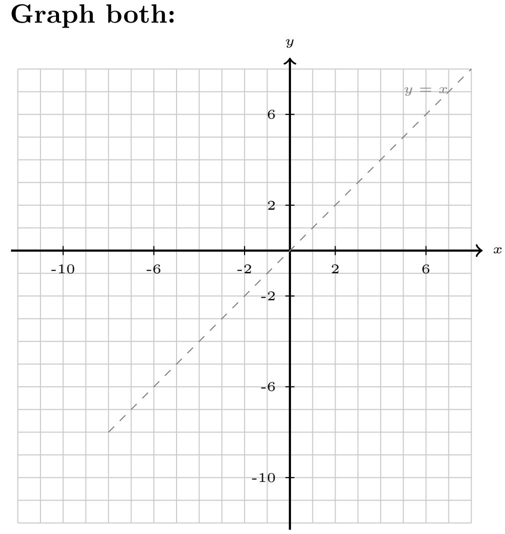
$f^{-1}(x) =$ ___________________

Key points: Start with parent, shift down 3

Parent	$f(x)$
$(-2, -8)$	$(-2,\ \ )$
$(-1, -1)$	$(-1,\ \ )$
$(0, 0)$	$(0,\ \ )$
$(1, 1)$	$(1,\ \ )$
$(2, 8)$	$(2,\ \ )$

$f(x)$	$f^{-1}(x)$
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)
(,)	(,)

Graph both:



Cubic & Cube Root Inverses — Practice (page 2)

Name: ____________________

10. $f(x) = \sqrt[3]{x-1}$ *(Cube root shifted right 1)*

Find the inverse:

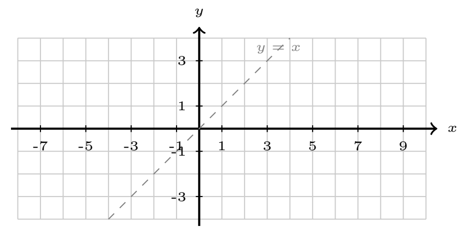
$f^{-1}(x) =$ ____________________

Key points: Start with parent, shift right 1

Parent	$f(x)$
$(-8,-2)$	$(\ \ ,-2)$
$(-1,-1)$	$(\ \ ,-1)$
$(0,0)$	$(\ \ ,0)$
$(1,1)$	$(\ \ ,1)$
$(8,2)$	$(\ \ ,2)$

$f(x)$	$f^{-1}(x)$
$(\ \ ,\ \ )$	$(\ \ ,\ \ )$
$(\ \ ,\ \ )$	$(\ \ ,\ \ )$
$(\ \ ,\ \ )$	$(\ \ ,\ \ )$
$(\ \ ,\ \ )$	$(\ \ ,\ \ )$
$(\ \ ,\ \ )$	$(\ \ ,\ \ )$

Graph both:



Part 3: True or False. Write T or F. If false, correct the statement.

11. ____ The inverse of $f(x) = x^3$ is $f^{-1}(x) = \sqrt[3]{x}$.

12. ____ If the point $(3, 27)$ is on $f(x) = x^3$, then the point $(3, 27)$ is on $f^{-1}(x)$.

13. ____ The graphs of a function and its inverse are reflections over the line $y = x$.

14. ____ To find an inverse, you swap x and y, then solve for x.

Part 4: Short Answer.

15. What line are $f(x)$ and $f^{-1}(x)$ reflected across?

Answer: ______________________________

16. A cubic function has been shifted up 6 units from the parent function. Write the equation of this function and its inverse.

$f(x) =$ ______________________ $f^{-1}(x) =$ ______________________

17. A cube root function has been shifted left 4 units from the parent function. Write the equation of this function and its inverse.

$f(x) =$ ______________________ $f^{-1}(x) =$ ______________________

Remember: • Inverse of x^3 is $\sqrt[3]{x}$ • Inverse of $\sqrt[3]{x}$ is x^3 • Points swap: $(a, b) \to (b, a)$ • Reflect over $y = x$